

Data Structures, Graphs, BFS, DFS: Exercise Handout with solutions

Data Structures and Algorithms (X_400614)

1. Would you use the stack or the queue data structure to implement the "Undo" feature in some word processor? What about for storing the mouse click events (such as copy-paste) when implementing the mouse user interface?

Undo - stack, mouse click - queue

2. You wish to create a database of stars. For each star, the database will store several megabytes of data. Considering that your database will store billions of stars, choose the data structure that will provide the best performance. With this data structure you should be able to find the smallest star, insert, and delete stars. Justify your choice.

[Heap. Takes less time to build compared to BST, the order is not important (which is the advantage of BST). The operations I need are still fast: $O(\log n)$, insert $(\log n)$ find min $O(1)$. And to build it I need $O(n)$ while I need $O(n \cdot \log n)$ for BST.]

3. Which of the following statements are true (n =number of vertices, m =number of edges in a graph):
 - a. BFS can be used to compute the connected components of an undirected graph in $O(m+n)$ time.
 - b. BFS can be used to compute the lengths of shortest path from starting vertex to every other vertex in the graph in $O(m+n)$ time, where shortest means the fewer number of edges.
 - c. DFS can be used to compute the strongly connected components of a directed acyclic graph in $O(m+n)$ time.

All of them are true (see book chapters introducing the algorithms).

4. Study the graph from Figure 1 and answer the questions a through e.

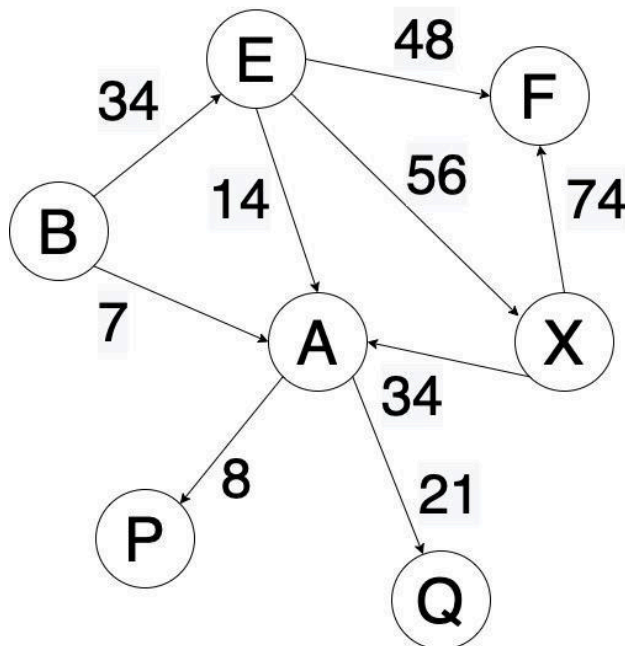


Figure 1: Directed graph with weighted edges.

- a) Create the adjacency matrix representation of the graph from Figure 1. Use -1 to represent that there is no edge and use the weights for the edges that exist.

-	B	E	A	F	X	P	Q
B		34	7				
E			14	48	56		
A						8	21
F							
F							
X			39	74			
P							
Q							

-1 elsewhere.

- b) Give one reason why the adjacency list might be a better representation compared to the adjacency matrix.
For sparse graph with not many edges, the list takes less memory space.
- c) Give one reason why the adjacency matrix might be better compared to the adjacency list representation.
Better running time of find/delete/insert an edge
- d) How would the matrix change if the graph was *undirected*?
It would be symmetric across the diagonal
- e) Perform a topological sort of the graph and show the final topological ordering. If more than one order exists, listing just one is sufficient.
BEXAFPQ or BEXAFQP or BEXFAQP or...

5. Study the graph from Figure 2 and answer the questions a and b.

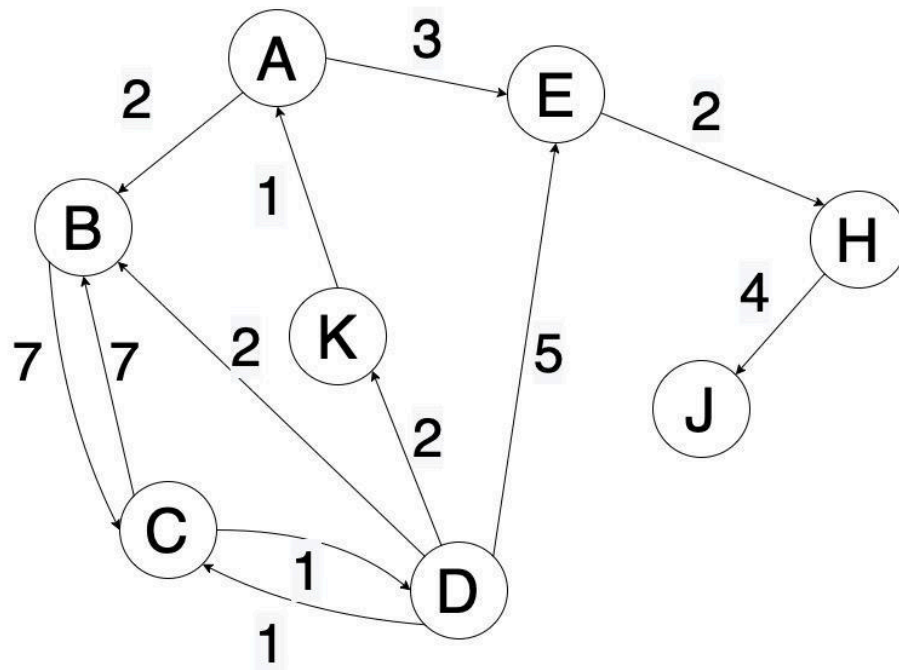


Figure 2: Directed graph with weighted edges.

- a) Given this graph, simulate the breadth first graph search algorithm starting with vertex A. Write down one possible order of vertices that would be processed by the BFS traversal.

ABECHDJK or AEBHCJDK

- b) Given this graph, simulate the depth first graph search algorithm starting with vertex A. Write down one possible order of vertices that would be processed by the DFS traversal.

AEHJBCKD